Q20 Ans: A

Power dissipated, $P = \frac{V^2}{R}$

Since the resistors are identical, $P \propto V^2$

R and Q are connected in parallel, hence

$$V_S : V_R : V_Q \quad \text{or} \quad {V_S}^2 : {V_R}^2 : {V_Q}^2$$
$$2 : 1 : 1 \quad \text{or} \quad 4 : 1 : 1$$

Power dissipated across R = $\frac{1}{6} \times 12$ = 2 W

Q21 Ans: D